

In these notes we reproduce the derivation in Dirac's 1938 paper on the classical theory of the radiating electron. The background of the paper is a challenge to the inconveniences of the Abraham–Lorentz electron model (spherically symmetric, finite size). Dirac abandons both "the electron's mass is entirely electromagnetic" and the finite-size model, treating the electron as a genuine point particle and letting Maxwell's theory hold all the way down to the singularity. The resulting infinite self-energy is not cured by modifying electrodynamics; instead it is subtracted off and absorbed into the observed mass (precisely the precursor of the renormalization idea later developed in quantum electrodynamics). Within this framework, using energy–momentum conservation, he derives a mathematically self-consistent, relativistically covariant equation of motion with radiation reaction for this point electron.

1. The "actual field" near the electron

In Maxwell theory the field is the solution of the two equations

$$\partial_\mu A^\mu = 0 \quad (1)$$

$$\square A^\mu = \frac{4\pi}{c} J^\mu \quad (2)$$

where (1) is the homogeneous equation—any linear combination of homogeneous solutions still satisfies (1)—and (2) is the sourced equation, whose two particular solutions were given in Jackson 14.1 (the Liénard–Wiechert potentials), namely the retarded and advanced solutions

$$A_{\text{ret}}^\mu = \frac{eV^\mu(\tau)}{V^\alpha [x_\alpha - z_\alpha(\tau)]} \Big|_{\tau=\tau_{\text{ret}}}$$

$$A_{\text{adv}}^\mu = \frac{eV^\mu(\tau)}{V^\alpha [x_\alpha - z_\alpha(\tau)]} \Big|_{\tau=\tau_{\text{adv}}}$$

Here τ is the proper time, $z(\tau)$ is the electron's trajectory, V is the electron's 4-velocity, and x is the spacetime coordinate of the observation point. $\tau_{\text{ret}}, \tau_{\text{adv}}$ are the two solutions (in the past and in the future) of the light-cone condition

$$[x^\alpha - z^\alpha(\tau)][x_\alpha - z_\alpha(\tau)] = 0$$

Dirac here does not simply discard the acausal advanced solution; instead he also counts it into the "actual field".

He argues that, since the "actual field" around the electron must satisfy both (1) and (2), it can necessarily be written as

$$A_{\text{act}}^\mu = \alpha A_{\text{ret}}^\mu + (1 - \alpha) A_{\text{adv}}^\mu + A_{\text{hom}}^\mu$$

where the first two terms are a linear combination of the two sourced particular solutions (to meet the sourced condition the coefficients must sum to 1), and the third term is a homogeneous solution. This expression can also be written in two other forms

$$A_{\text{act}}^\mu = A_{\text{ret}}^\mu + \overbrace{(1 - \alpha)(A_{\text{adv}}^\mu - A_{\text{ret}}^\mu) + A_{\text{hom}}^\mu}^{A_{\text{in}}^\mu}$$

$$= A_{\text{adv}}^\mu + \underbrace{\alpha(A_{\text{ret}}^\mu - A_{\text{adv}}^\mu) + A_{\text{hom}}^\mu}_{A_{\text{out}}^\mu}$$

where $A_{\text{in}}, A_{\text{out}}$ are two recombined homogeneous (sourceless) solutions. For now we treat them merely as a mathematical construction and defer their physical meaning; the "in/out" labels are likewise not over-interpreted yet.

Using

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

to write A as a field-strength tensor, the above becomes

$$F_{\text{act}}^{\mu\nu} = F_{\text{ret}}^{\mu\nu} + F_{\text{in}}^{\mu\nu} = F_{\text{adv}}^{\mu\nu} + F_{\text{out}}^{\mu\nu}$$

Introduce the new quantities

$$F_{\text{rad}}^{\mu\nu} = F_{\text{out}}^{\mu\nu} - F_{\text{in}}^{\mu\nu} = F_{\text{ret}}^{\mu\nu} - F_{\text{adv}}^{\mu\nu} \quad (3)$$

$$f^{\mu\nu} = F_{\text{in}}^{\mu\nu} + \frac{1}{2}(F_{\text{ret}}^{\mu\nu} - F_{\text{adv}}^{\mu\nu}) \quad (4)$$

Again, temporarily ignore the "rad" label and treat it as a mathematical construction. The "actual field" F_{act} can then be written as

$$F_{\text{act}}^{\mu\nu} = f^{\mu\nu} + \frac{1}{2}(F_{\text{ret}}^{\mu\nu} + F_{\text{adv}}^{\mu\nu}) \quad (5)$$

2. Taylor expansion of the retarded and advanced solutions near the worldline

In Jackson 14.1 the field-strength tensors of the retarded and advanced solutions were given (eq (14.11))—where the advanced solution needs an overall minus sign, which comes from the θ function in (14.10))

$$F_{\text{ret}}^{\mu\nu} = \frac{e}{R^\alpha V_\alpha} \frac{d}{d\tau} \left(\frac{R^\mu V^\nu - R^\nu V^\mu}{R^\alpha V_\alpha} \right) \Big|_{\tau_{\text{ret}}} \quad R^\mu \equiv x^\mu - z^\mu(\tau) \quad (6)$$

$$F_{\text{adv}}^{\mu\nu} = -\frac{e}{R^\alpha V_\alpha} \frac{d}{d\tau} \left(\frac{R^\mu V^\nu - R^\nu V^\mu}{R^\alpha V_\alpha} \right) \Big|_{\tau_{\text{adv}}} \quad (7)$$

To simplify, set $c = 1$ below.

Here we Taylor-expand (6) and (7) near the electron's worldline. Specifically, for any observation point x , let τ_0 be the proper time at which the worldline $z(\tau)$ is closest to x , i.e.

$$\frac{d[x^\mu - z^\mu(\tau)][x_\mu - z_\mu(\tau)]}{d\tau} \Big|_{\tau_0} = 0 \quad \Rightarrow \quad V^\mu(\tau_0) \overbrace{[x_\mu - z_\mu(\tau_0)]}^{\equiv \gamma_\mu} = 0$$

Applying this orthogonality relation in the electron's instantaneous rest frame fixes the time component of γ^μ to zero; hence its 4-norm is negative (spacelike), denoted

$$\gamma^\mu \gamma_\mu = -\epsilon^2 < 0$$

Replacing the proper-time variable τ by the shift σ relative to τ_0 ,

$$\tau = \tau_0 - \sigma$$

we expand $R(\tau), V(\tau)$ near τ_0 :

$$\begin{aligned} R^\mu(\tau) &= x^\mu - z^\mu(\tau) = \gamma^\mu + z^\mu(\tau_0) - \left[z^\mu(\tau_0) - V^\mu \sigma + \frac{\dot{V}^\mu}{2} \sigma^2 - \frac{\ddot{V}^\mu}{6} \sigma^3 + \dots \right] \\ &= \gamma^\mu + V^\mu \sigma - \frac{\dot{V}^\mu}{2} \sigma^2 + \frac{\ddot{V}^\mu}{6} \sigma^3 + \dots \end{aligned} \quad (8)$$

$$V^\mu(\tau) = V^\mu - \dot{V}^\mu \sigma + \frac{\ddot{V}^\mu}{2} \sigma^2 + \dots \quad (9)$$

where the V^μ on the right is the 4-velocity at τ_0 . Near the worldline $|\gamma| \sim O(\epsilon)$ and $\sigma \sim O(\epsilon)$, so all order-of-magnitude analysis below is measured in $O(\epsilon^n)$; e.g. $\gamma \sigma^3 \sim O(\epsilon^4)$.

With the notation

$$A^{[\alpha} B^{\beta]} = A^\alpha B^\beta - A^\beta B^\alpha$$

the relevant pieces of (6) become, by (8) and (9),

$$\begin{aligned} R^\mu V^\nu - R^\nu V^\mu &= \gamma^{[\mu} V^{\nu]} + \left(\overbrace{V^{[\mu} V^{\nu]}}^0 - \gamma^{[\mu} \dot{V}^{\nu]} \right) \sigma + \left(\frac{\gamma^{[\mu} \ddot{V}^{\nu]}}{2} - V^{[\mu} \dot{V}^{\nu]} - \frac{\overbrace{\dot{V}^{[\mu} V^{\nu]}}^{+V^{[\mu} \dot{V}^{\nu]}/2}}{2} \right) \sigma^2 + \\ &\quad \left(\frac{\ddot{V}^{[\mu} V^{\nu]}}{6} + \underbrace{\frac{\dot{V}^{[\mu} \dot{V}^{\nu]}}{2}}_0 + \underbrace{\frac{V^{[\mu} \ddot{V}^{\nu]}}{2}}_{-\ddot{V}^{[\mu} V^{\nu]}/2} \right) \sigma^3 + \dots \\ &= \gamma^{[\mu} V^{\nu]} - \gamma^{[\mu} \dot{V}^{\nu]} \sigma + \left(\frac{\gamma^{[\mu} \ddot{V}^{\nu]}}{2} - \frac{V^{[\mu} \dot{V}^{\nu]}}{2} \right) \sigma^2 - \frac{\dot{V}^{[\mu} V^{\nu]}}{3} \sigma^3 + \dots \\ R^\alpha V_\alpha &= \overbrace{\gamma^\alpha V_\alpha}^0 + \left(\overbrace{V^\alpha V_\alpha}^1 - \gamma^\alpha \dot{V}_\alpha \right) \sigma + \left(\frac{\gamma^\alpha \ddot{V}_\alpha}{2} - \frac{\overbrace{3V^\alpha \dot{V}_\alpha}^0}{2} \right) \sigma^2 + \left(\frac{\dot{V}^\alpha \dot{V}_\alpha}{2} + \frac{\overbrace{-2\dot{V}^\alpha \ddot{V}_\alpha/3}}{3} \right) \sigma^3 + \dots \\ &= \left(\underbrace{1 - \gamma \dot{V}}_{\equiv \kappa} \right) \sigma + \frac{\gamma \ddot{V}}{2} \sigma^2 - \frac{\dot{V}^2}{6} \sigma^3 + \dots \end{aligned}$$

Therefore

$$\frac{1}{R^\alpha V_\alpha} = \frac{1}{\kappa \sigma} \left(1 + \frac{\gamma \ddot{V}}{2\kappa} \sigma - \frac{\dot{V}^2}{6\kappa} \sigma^2 + \dots \right)^{-1} \approx \frac{1}{\kappa \sigma} \left(1 - \frac{\gamma \ddot{V}}{2} \sigma + \frac{\dot{V}^2}{6} \sigma^2 \right) \quad (10)$$

Note $\kappa \sim 1 + O(\epsilon)$, so to $O(\epsilon^2)$ accuracy inside the parentheses κ may be dropped. Hence

$$\frac{R^\mu V^\nu - R^\nu V^\mu}{R^\alpha V_\alpha} \approx \frac{1}{\kappa \sigma} \left[\gamma^{[\mu} V^{\nu]} - \gamma^{[\mu} \dot{V}^{\nu]} \sigma - \frac{(\gamma \ddot{V}) \gamma^{[\mu} V^{\nu]}}{2} \sigma + \frac{\dot{V}^2 \gamma^{[\mu} V^{\nu]}}{6} \sigma^2 + \frac{\gamma^{[\mu} \ddot{V}^{\nu]}}{2} \sigma^2 - \frac{V^{[\mu} \dot{V}^{\nu]}}{2} \sigma^2 - \frac{\dot{V}^{[\mu} V^{\nu]}}{3} \sigma^3 \right]$$

The $d/d\tau$ in (6), (7) can be replaced by $-d/d\sigma$:

$$\frac{d}{d\tau} \left(\frac{R^\mu V^\nu - R^\nu V^\mu}{R^\alpha V_\alpha} \right) \approx -\frac{1}{\kappa} \left(-\gamma^{[\mu} V^{\nu]} \sigma^{-2} + \frac{\dot{V}^2 \gamma^{[\mu} V^{\nu]}}{6} + \frac{\gamma^{[\mu} \ddot{V}^{\nu]}}{2} - \frac{V^{[\mu} \dot{V}^{\nu]}}{2} - \frac{2\dot{V}^{[\mu} V^{\nu]}}{3} \sigma \right)$$

Multiplying by (6), (7) gives

$$F_{\text{ret|adv}}^{\mu\nu} = \pm \frac{e}{\kappa^2} \left[\overbrace{\gamma^{[\mu} V^{\nu]}}^G \sigma^{-3} - \frac{(\gamma \ddot{V}) \gamma^{[\mu} V^{\nu]}}{2} \sigma^{-2} - \frac{\overbrace{\gamma^{[\mu} \ddot{V}^{\nu]}}^H}{2} \sigma^{-1} + \frac{\overbrace{V^{[\mu} \dot{V}^{\nu]}}^J}{2} \sigma^{-1} + \frac{2\overbrace{\dot{V}^{[\mu} V^{\nu]}}^K}{3} \right] \sigma_{\text{ret|adv}} \quad (11)$$

This must be evaluated at the corresponding retarded/advanced instants $\sigma_{\text{ret}}/\sigma_{\text{adv}}$. We now solve for them.

In the electron's rest frame, x and $z(\tau_0)$ are simultaneous, while τ_{ret} (τ_{adv}) must occur before (after) τ_0 (the ordering is frame-independent), so we write

$$\tau_{\text{ret}} = \tau_0 - \sigma_{\text{ret}} \quad \sigma_{\text{ret}} > 0$$

and similarly

$$\tau_{\text{adv}} = \tau_0 - \sigma_{\text{adv}} \quad \sigma_{\text{adv}} < 0$$

Imposing the light-cone condition that defines $\tau_{\text{ret}}/\tau_{\text{adv}}$,

$$\begin{aligned} 0 &= R^\mu R_\mu = \overbrace{\gamma^\mu \gamma_\mu}^{-\epsilon^2} + 2\overbrace{V^\mu \gamma_\mu}^0 \sigma + \left(\overbrace{V^\mu V_\mu}^1 - \gamma^\mu \dot{V}_\mu \right) \sigma^2 + \left(-\overbrace{V^\mu \dot{V}_\mu}^0 + \frac{1}{3} \gamma^\mu \ddot{V}_\mu \right) \sigma^3 + \left(\frac{\dot{V}^\mu \dot{V}_\mu}{4} + \frac{\overbrace{V^\mu \ddot{V}_\mu}^{-\dot{V}^\mu \dot{V}_\mu/3}}{3} \right) \sigma^4 + \dots \\ &= -\epsilon^2 + \kappa \sigma^2 + \frac{\gamma \ddot{V}}{3} \sigma^3 - \frac{\dot{V}^2}{12} \sigma^4 + \dots \end{aligned} \quad (12)$$

At zeroth order we may take

$$\sigma^{(0)} \approx \pm \epsilon \quad + \text{ for retarded, } - \text{ for advanced}$$

Substituting the zeroth-order value back into the σ^3, σ^4 terms of (12), the first-order equation is

$$-\epsilon^2 + \kappa \sigma^2 \pm \frac{\gamma \ddot{V}}{3} \epsilon^3 - \frac{\dot{V}^2}{12} \epsilon^4 = 0$$

giving the approximate solution

$$\sigma \approx \pm \frac{\epsilon}{\sqrt{\kappa}} \left(1 \mp \frac{\gamma \ddot{V}}{6} \epsilon + \frac{\dot{V}^2}{24} \epsilon^2 \right)$$

Substituting σ back into (11), one must carefully account for all orders $O(\epsilon^k)$:

$$\begin{aligned} \sigma^{-3} &\approx \pm \kappa^{3/2} \epsilon^{-3} \left(1 \pm \frac{\gamma \ddot{V}}{2} \epsilon - \frac{\dot{V}^2}{8} \epsilon^2 \right) \\ \sigma^{-2} &\approx \kappa \epsilon^{-2} \left(1 \pm \frac{\gamma \ddot{V}}{3} \epsilon - \frac{\dot{V}^2}{12} \epsilon^2 \right) \\ \sigma^{-1} &\approx \pm \kappa^{1/2} \epsilon^{-1} \left(1 \pm \frac{\gamma \ddot{V}}{6} \epsilon - \frac{\dot{V}^2}{24} \epsilon^2 \right) \end{aligned}$$

Inside the bracket of (11), at $O(1)$ accuracy, the four terms other than K expand as

$$\begin{aligned}
G \sim O(\epsilon) : \quad & G\sigma^{-3} \approx \pm G\kappa^{3/2}\epsilon^{-3} + \frac{(\gamma\ddot{V})G}{2}\kappa^{3/2}\epsilon^{-2} \mp \frac{\dot{V}^2 G}{8}\kappa^{3/2}\epsilon^{-1} \\
(\gamma\ddot{V})G \sim O(\epsilon^2) : \quad & \frac{(\gamma\ddot{V})G}{2}\sigma^{-2} \approx \frac{(\gamma\ddot{V})G}{2}\kappa\epsilon^{-2} \\
H \sim O(\epsilon) : \quad & \frac{H}{2}\sigma^{-1} \approx \pm \frac{H}{2}\kappa^{1/2}\epsilon^{-1} \\
J \sim O(1) : \quad & \frac{J}{2}\sigma^{-1} \approx \pm \frac{J}{2}\kappa^{1/2}\epsilon^{-1}
\end{aligned}$$

so that (11) becomes

$$F_{\text{ret|adv}}^{\mu\nu} \approx \pm e\kappa^{-1/2} \left[\pm G\epsilon^{-3} + \frac{(\gamma\ddot{V})G}{2}\epsilon^{-2} \mp \frac{\dot{V}^2 G}{8}\epsilon^{-1} - \frac{(\gamma\ddot{V})G}{2}\kappa^{-1/2}\epsilon^{-2} \mp \frac{H}{2}\kappa^{-1}\epsilon^{-1} \pm \frac{J}{2}\kappa^{-1}\epsilon^{-1} + \frac{2K}{3}\kappa^{-3/2} \right]$$

Inside the bracket, at $O(1)$ accuracy, all κ except in the J term may be set to 1, while in the J term κ must be kept to $O(\epsilon)$. The ϵ^{-2} terms then cancel, and the divergent terms carry only odd powers, i.e.

$$F_{\text{ret|adv}}^{\mu\nu} \approx e\kappa^{-1/2} \left[\gamma^{[\mu} V^{\nu]} \epsilon^{-3} - \frac{\dot{V}^2 \gamma^{[\mu} V^{\nu]}}{8} \epsilon^{-1} - \frac{\gamma^{[\mu} \ddot{V}^{\nu]}}{2} \epsilon^{-1} + \frac{V^{[\mu} \dot{V}^{\nu]}}{2} (1 + \gamma\dot{V}) \epsilon^{-1} \pm \frac{2\ddot{V}^{[\mu} V^{\nu]}}{3} \right] \quad (13)$$

This is now very interesting: we know that $F_{\text{ret}}, F_{\text{adv}}$ diverge at the electron's location, but the only diverging powers are odd, and the divergent terms of the retarded and advanced solutions carry the same sign. So there is great motivation to subtract one from the other, so that the divergence cancels.

That is precisely why we define f in (4). We shall see that the equation of motion of f closely resembles Newton's law, and through it the renormalized electron mass is redefined. Substituting (13) into (5), we compute the field-strength tensor of the "actual field":

$$\begin{aligned}
F_{\text{act}}^{\mu\nu} &= f^{\mu\nu} + \frac{1}{2} (F_{\text{ret}}^{\mu\nu} + F_{\text{adv}}^{\mu\nu}) \\
&= f^{\mu\nu} + e\kappa^{-1/2} \left[\left(\epsilon^{-3} - \frac{\dot{V}^2 \epsilon^{-1}}{8} \right) \gamma^{[\mu} V^{\nu]} + \frac{(1 + \gamma\dot{V})\epsilon^{-1}}{2} V^{[\mu} \dot{V}^{\nu]} - \frac{\epsilon^{-1}}{2} \gamma^{[\mu} \ddot{V}^{\nu]} \right] \quad (14)
\end{aligned}$$

3. The 4-momentum on the worldline tube

Dirac obtains the equation of motion of the tensor f by computing the conservation of the "actual field" 4-momentum on a tube surrounding the worldline. Let us read through this argument in detail. First we define the tube: it is a 3D hypersurface built from a 2D spacelike surface defined at each proper time τ ,

$$x^\mu(\tau) \equiv z^\mu(\tau) + \gamma^\mu(\tau) \quad \text{where } \gamma^\mu \gamma_\mu = -\epsilon^2 \text{ and } \gamma^\mu V_\mu = 0 \quad (15)$$

swept out as τ runs. In the electron's instantaneous rest frame this 2D surface is a sphere of radius ϵ centered on the electron. This tube wall, together with the two end caps at the initial and final proper times τ_1, τ_2 , forms a closed 4-dimensional spacetime region V . Gauss's theorem on this region reads

$$\int_V \partial_\nu \Theta^{\mu\nu} d^4x = \int_{\partial V} \Theta^{\mu\nu} d^3a_\nu = \int_{\text{top cap}} \Theta^{\mu\nu} d^3a_\nu + \int_{\text{bottom cap}} \Theta^{\mu\nu} d^3a_\nu + \int_{\text{tube wall}} \Theta^{\mu\nu} d^3a_\nu \quad (16)$$

From here Dirac gives a somewhat vague description to argue that the flux through the tube wall must be a perfect differential.

"We surround the singular world-line in space-time by a thin tube, whose radius is much smaller than any length of physical importance in our one-electron problem, and we calculate the flow of energy and momentum across the (three-dimensional) surface of this tube, using the stress tensor $T_{\mu\rho}$ of Maxwell's theory, calculated from the actual field $F_{\text{act}} \dots$ We then have the information that the total flow of energy (or momentum) out from the surface of any finite length of tube must equal the difference in the energy (or momentum) residing within the tube at the two ends of this length, and must thus depend only on conditions at the two ends of this length. In mathematical language, the rate of flow of energy (or momentum) out from the surface of the tube must be a perfect differential."

In my view this passage is less an argument than an implicit assumption, since the conclusion is not really derived.

His conclusion is that

$$\int_{\text{tube wall}} \Theta^{\mu\nu} d^3 a_\nu = Q^\mu(\tau_2) - Q^\mu(\tau_1)$$

where Q is some physical quantity on the two end caps at proper times τ_1, τ_2 . But from (16), his conclusion holds only if the left-hand 4-volume integral can be written purely as the difference, at τ_1 and τ_2 , of some end-cap quantity. Granted, by the usual electromagnetic theory the integrand on the left is the Lorentz force on the electron; but he cannot then use that result to derive the corrected equation of motion later on—that would be circular reasoning.

The only logical way out that makes his later theory work is to admit that his "perfect differential" conclusion is an added assumption.

First we compute the surface element $d^3 a_\nu$, defined by (15). In the electron's rest frame, a point x on the surface (15) can be parametrized as

$$x^\mu(\tau, \phi, \theta) = z^\mu(\tau) + \epsilon n^\mu(\tau) \quad n^\mu(\tau) = (0, \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z})$$

Note that the parameter ordering here is τ - ϕ - θ ; this choice makes the positive direction of n point away from the worldline, i.e. the determinant

$$\det \begin{bmatrix} n^0 & \partial_\tau x^0 & \partial_\phi x^0 & \partial_\theta x^0 \\ n^1 & \partial_\tau x^1 & \partial_\phi x^1 & \partial_\theta x^1 \\ n^2 & \partial_\tau x^2 & \partial_\phi x^2 & \partial_\theta x^2 \\ n^3 & \partial_\tau x^3 & \partial_\phi x^3 & \partial_\theta x^3 \end{bmatrix} = \det \begin{bmatrix} 0 & 1 + \epsilon \dot{n}^0 & 0 & 0 \\ \sin \theta \cos \phi & \epsilon \dot{n}^1 & -\epsilon \sin \theta \sin \phi & \epsilon \cos \theta \cos \phi \\ \sin \theta \sin \phi & \epsilon \dot{n}^2 & \epsilon \sin \theta \cos \phi & \epsilon \cos \theta \sin \phi \\ \cos \theta & \epsilon \dot{n}^3 & 0 & -\epsilon \sin \theta \end{bmatrix} \\ = (1 + \epsilon \dot{n}^0) \epsilon^2 \sin \theta > 0$$

If the parameter parity changes, the definitions of "outward" and "inward" are flipped. With the parametrization, the surface element is computed via

$$d^3 a_\nu = \epsilon_{\nu\mu\rho\lambda} \frac{\partial x^\mu}{\partial \tau} \frac{\partial x^\rho}{\partial \phi} \frac{\partial x^\lambda}{\partial \theta} d\tau d\phi d\theta$$

and the determinant was just computed. The time derivative $\dot{n}^0$ is obtained by going to the rest frame together with the covariance of the inner product,

$$\dot{n}^0 = \dot{n}^\alpha V_\alpha = -n^\alpha \dot{V}_\alpha \quad \implies \quad 1 + \epsilon \dot{n}^0 = 1 - \gamma \dot{V} = \kappa$$

In the rest frame the component form of the surface element is

$$d^3 a_\nu = \kappa \epsilon^2 \sin \theta d\theta d\phi d\tau \times (0, \sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

With the metric convention $g = (+---)$, writing it as a covariant (lower-index) tensor requires a sign, i.e.

$$d^3 a_\nu = -\kappa \epsilon^2 n_\nu d\tau \sin \theta d\theta d\phi = -\kappa \epsilon \gamma_\nu d\tau d\Omega \quad (17)$$

We now compute the flux through the tube wall. The stress tensor is given by Jackson (12.113),

$$\Theta^{\mu\rho} = \frac{1}{4\pi} \left(g^{\mu\lambda} F_{\lambda\nu} F^{\nu\rho} + \frac{1}{4} g^{\mu\rho} F_{\alpha\beta} F^{\alpha\beta} \right) = \frac{1}{4\pi} \left(F^\mu{}_\nu F^{\nu\rho} + \frac{1}{4} g^{\mu\rho} F^{\alpha\beta} F_{\alpha\beta} \right)$$

where F is the actual field F_{act} . By (17), the differential flux through the surface element is

$$\Theta^{\mu\rho} d^3 a_\rho = -\frac{\kappa \epsilon}{4\pi} \left(F^\mu{}_\nu F^{\nu\rho} \gamma_\rho + \frac{1}{4} \gamma^\mu F_{\alpha\beta} F^{\alpha\beta} \right) d\tau d\Omega \quad (18)$$

First, since the above will ultimately be multiplied by $O(|\gamma|\epsilon) = O(\epsilon^2)$, as $\epsilon \rightarrow 0$ we only need to keep the FF terms to $O(\epsilon^{-2})$ accuracy.

By (14), for the first FF term, the cross terms within $O(\epsilon^{-2})$ accuracy are

$$\begin{aligned} (\gamma^{[\mu} V_{\nu]}) (\gamma^{[\nu} V^{\rho]}) &= (\gamma^\mu V_\nu - \gamma_\nu V^\mu) (\gamma^\nu V^\rho - \gamma^\rho V^\nu) = -\gamma^\mu \gamma^\rho + \epsilon^2 V^\mu V^\rho \\ (\gamma^{[\mu} V_{\nu]}) (V^{[\nu} \dot{V}^{\rho]}) &= (\gamma^\mu V_\nu - \gamma_\nu V^\mu) (V^\nu \dot{V}^\rho - V^\rho \dot{V}^\nu) = \gamma^\mu \dot{V}^\rho + (\gamma \dot{V}) V^\mu V^\rho \\ (\gamma^{[\nu} V^{\rho]}) (V^{[\mu} \dot{V}_\nu]) &= (\gamma^\nu V^\rho - \gamma^\rho V^\nu) (V^\mu \dot{V}_\nu - V_\nu \dot{V}^\mu) = \dot{V}^\mu \gamma^\rho + (\gamma \dot{V}) V^\mu V^\rho \\ (\gamma^{[\mu} V_{\nu]}) (\gamma^{[\nu} \dot{V}^{\rho]}) &= (\gamma^\mu V_\nu - \gamma_\nu V^\mu) (\gamma^\nu \dot{V}^\rho - \gamma^\rho \dot{V}^\nu) = \dot{V}^2 \gamma^\mu \gamma^\rho + \epsilon^2 V^\mu \dot{V}^\rho + (\gamma \ddot{V}) V^\mu \gamma^\rho \\ (\gamma^{[\nu} V^{\rho]}) (\gamma^{[\mu} \dot{V}_\nu]) &= (\gamma^\nu V^\rho - \gamma^\rho V^\nu) (\gamma^\mu \dot{V}_\nu - \gamma_\nu \dot{V}^\mu) = \dot{V}^2 \gamma^\mu \gamma^\rho + \epsilon^2 \dot{V}^\mu V^\rho + (\gamma \ddot{V}) \gamma^\mu V^\rho \\ (V^{[\mu} \dot{V}_\nu]) (V^{[\nu} \dot{V}^{\rho]}) &= (V^\mu \dot{V}_\nu - V_\nu \dot{V}^\mu) (V^\nu \dot{V}^\rho - V^\rho \dot{V}^\nu) = -\dot{V}^2 V^\mu V^\rho - \dot{V}^\mu \dot{V}^\rho \end{aligned}$$

hence

$$\begin{aligned}
F^\mu{}_\nu F^{\nu\rho} &= \frac{e^2}{\kappa} \left\{ \left(\epsilon^{-6} - \frac{\dot{V}^2 \epsilon^{-4}}{4} \right) (-\gamma^\mu \gamma^\rho + \epsilon^2 V^\mu V^\rho) + \frac{(1 + \gamma \dot{V}) \epsilon^{-4}}{2} [\gamma^\mu \dot{V}^\rho + \dot{V}^\mu \gamma^\rho + 2(\gamma \dot{V}) V^\mu V^\rho] - \right. \\
&\quad \left. \frac{\epsilon^{-4}}{2} [(\gamma \ddot{V})(V^\mu \gamma^\rho + \gamma^\mu V^\rho) + \epsilon^2 (V^\mu \ddot{V}^\rho + \ddot{V}^\mu V^\rho) + 2\dot{V}^2 \gamma^\mu \gamma^\rho] - \frac{\epsilon^{-2}}{4} (\dot{V}^2 V^\mu V^\rho + \dot{V}^\mu \dot{V}^\rho) \right\} + \\
&\quad e \epsilon^{-3} (f^\mu{}_\nu \gamma^{[\nu} V^{\rho]} + f^{\nu\rho} \gamma^{[\mu} V_{\nu]}) \\
F^\mu{}_\nu F^{\nu\rho} \gamma_\rho &= \frac{e^2}{\kappa} \left\{ \left(\epsilon^{-6} - \frac{\dot{V}^2 \epsilon^{-4}}{4} \right) \epsilon^2 \gamma^\mu + \frac{(1 + \gamma \dot{V}) \epsilon^{-4}}{2} [(\gamma \dot{V}) \gamma^\mu - \epsilon^2 \dot{V}^\mu] + \epsilon^{-2} \dot{V}^2 \gamma^\mu - \frac{\epsilon^{-2}}{4} (\gamma \dot{V}) \dot{V}^\mu \right\} + \\
&\quad e \epsilon^{-3} (\epsilon^2 f^\mu{}_\nu V^\nu + \gamma^\mu f^{\nu\rho} V_\nu \gamma_\rho)
\end{aligned} \tag{19}$$

For the second FF term, the useful cross terms are

$$\begin{aligned}
(\gamma^{[\alpha} V^{\beta]}) (\gamma_{[\alpha} V_{\beta]}) &= (\gamma^\alpha V^\beta - \gamma^\beta V^\alpha) (\gamma_\alpha V_\beta - \gamma_\beta V_\alpha) = -2\epsilon^2 \\
(\gamma^{[\alpha} V^{\beta]}) (V_{[\alpha} \dot{V}_{\beta]}) &= (\gamma^\alpha V^\beta - \gamma^\beta V^\alpha) (V_\alpha \dot{V}_\beta - V_\beta \dot{V}_\alpha) = -2(\gamma \dot{V}) \\
(\gamma_{[\alpha} V_{\beta]}) (V^{[\alpha} \dot{V}^{\beta]}) &= (\gamma_\alpha V_\beta - \gamma_\beta V_\alpha) (V^\alpha \dot{V}^\beta - V^\beta \dot{V}^\alpha) = -2(\gamma \dot{V}) \\
(\gamma^{[\alpha} V^{\beta]}) (\gamma_{[\alpha} \ddot{V}_{\beta]}) &= (\gamma^\alpha V^\beta - \gamma^\beta V^\alpha) (\gamma_\alpha \ddot{V}_\beta - \gamma_\beta \ddot{V}_\alpha) = 2\epsilon^2 \dot{V}^2 \\
(\gamma_{[\alpha} V_{\beta]}) (\gamma^{[\alpha} \ddot{V}^{\beta]}) &= (\gamma_\alpha V_\beta - \gamma_\beta V_\alpha) (\gamma^\alpha \ddot{V}^\beta - \gamma^\beta \ddot{V}^\alpha) = 2\epsilon^2 \dot{V}^2 \\
(V^{[\alpha} \dot{V}^{\beta]}) (V_{[\alpha} \dot{V}_{\beta]}) &= (V^\alpha \dot{V}^\beta - V^\beta \dot{V}^\alpha) (V_\alpha \dot{V}_\beta - V_\beta \dot{V}_\alpha) = 2\dot{V}^2
\end{aligned}$$

so that

$$\begin{aligned}
F^{\alpha\beta} F_{\alpha\beta} &= \frac{e^2}{\kappa} \left\{ \left(\epsilon^{-6} - \frac{\dot{V}^2 \epsilon^{-4}}{4} \right) (-2\epsilon^2) - 2(1 + \gamma \dot{V})(\gamma \dot{V}) \epsilon^{-4} - 2\dot{V}^2 \epsilon^{-2} + \frac{\dot{V}^2}{2} \epsilon^{-2} \right\} + e \epsilon^{-3} (f_{\alpha\beta} \gamma^{[\alpha} V^{\beta]} + f^{\alpha\beta} \gamma_{[\alpha} V_{\beta]}) \\
&= \frac{e^2}{\kappa} \left\{ -2[1 + \gamma \dot{V} + (\gamma \dot{V})^2] \epsilon^{-4} - \dot{V}^2 \epsilon^{-2} \right\} + 4e \epsilon^{-3} f_{\alpha\beta} \gamma^\alpha V^\beta \\
\frac{1}{4} \gamma^\mu F_{\alpha\beta} F^{\alpha\beta} &= \frac{e^2}{\kappa} \left\{ -\frac{1}{2} [1 + \gamma \dot{V} + (\gamma \dot{V})^2] \epsilon^{-4} - \frac{\dot{V}^2}{4} \epsilon^{-2} \right\} \gamma^\mu + e \epsilon^{-3} \gamma^\mu f_{\alpha\beta} \gamma^\alpha V^\beta
\end{aligned} \tag{20}$$

With the f tensor being antisymmetric,

$$f^{\nu\rho} V_\nu \gamma_\rho = -f_{\alpha\beta} \gamma^\alpha V^\beta$$

adding (19) and (20) gives

$$F^\mu{}_\nu F^{\nu\rho} \gamma_\rho + \frac{1}{4} \gamma^\mu F_{\alpha\beta} F^{\alpha\beta} = \frac{e^2}{\kappa} \left[\left(\frac{1}{2} \epsilon^{-4} + \frac{1}{2} \dot{V}^2 \epsilon^{-2} \right) \gamma^\mu - \frac{1}{2} \epsilon^{-2} \left(1 + \frac{3}{2} \gamma \dot{V} \right) \dot{V}^\mu \right] + e \epsilon^{-1} f^\mu{}_\nu V^\nu$$

Substituting into (18) and integrating,

$$\begin{aligned}
\int_{\text{tube wall}} \Theta^{\mu\rho} d^3 a_\rho &= - \int \frac{\kappa \epsilon}{4\pi} \left\{ \frac{e^2}{\kappa} \left[\left(\frac{1}{2} \epsilon^{-4} + \frac{1}{2} \dot{V}^2 \epsilon^{-2} \right) \gamma^\mu - \frac{1}{2} \epsilon^{-2} \left(1 + \frac{3}{2} \gamma \dot{V} \right) \dot{V}^\mu \right] + e \epsilon^{-1} f^\mu{}_\nu V^\nu \right\} d\tau d\Omega \\
&= -\frac{1}{4\pi} \int \left\{ e^2 \left[\left(\frac{1}{2} \epsilon^{-3} + \frac{1}{2} \dot{V}^2 \epsilon^{-1} \right) \gamma^\mu - \frac{1}{2} \epsilon^{-1} \dot{V}^\mu - \frac{3}{4} (\gamma \dot{V}) \dot{V}^\mu \epsilon^{-1} \right] + (1 + \gamma \dot{V}) e f^\mu{}_\nu V^\nu \right\} d\tau d\Omega \\
&= \int \left(\frac{e^2}{2\epsilon} \dot{V}^\mu - e f^\mu{}_\nu V^\nu \right) d\tau
\end{aligned}$$

where in the last step, by spherical symmetry, all terms carrying γ integrate to zero. Per the earlier assumption, this tube-wall flux integral is a quantity that depends only on the two end caps, so it can be written as

$$\frac{e^2}{2\epsilon} \dot{V}^\mu - e f^{\mu\nu} V_\nu = \dot{B}^\mu$$

Dirac says this is the information obtainable from the conservation law; for B we can fix only one constraint, namely

$$\dot{B}^\mu V_\mu = 0$$

which follows from contracting the above with V ($\dot{V}^\mu V_\mu = 0$ and f is antisymmetric). The simplest choice for B is

$$B^\mu = k V^\mu$$

(there are other choices, but they involve higher derivatives of the velocity, too inelegant for something as simple as an electron). Under this assumption we obtain

$$ef^{\mu\nu}V_\nu = \left(\frac{e^2}{2\epsilon} - k \right) \dot{V}^\mu \quad (21)$$

4. Electron mass renormalization

The reading of (21) is now very interesting: it is precisely the covariant equation of motion of the electron, equating "force" with "mass $\times$ acceleration." The LHS is the Lorentz 4-force on the electron, where f is the singularity-free field left after subtracting from the actual field the mean of the retarded and advanced fields. The RHS has the form mass $\times$ acceleration. The mass consists of two parts: $e^2/2\epsilon$ is the electromagnetic mass carried by the electron's own Coulomb field, while k is the bare mass (non-electromagnetic, mechanical inertia) introduced through the definition of B —an undetermined constant. Individually neither is observable, and the electromagnetic part is infinite; but experiment can only measure their sum. We therefore redefine this combination as the observable physical mass, letting the divergent electromagnetic mass be absorbed by an equally divergent bare mass, leaving a finite observable m . This is the classical prototype of mass renormalization.

Now let us revisit the various vaguely-defined combinations (see (3) and (4)). Why "in/out"? If we regard the electron as a scattering center, "in" is the free field arriving from infinity before the interaction, and "out" is the field flying off after the interaction. As a physical source, the electron can only affect the field in the retarded way (causally, along the forward light cone), so the part the electron itself contributes is F_{ret} ; subtracting it from the actual field leaves the free incident field that has nothing to do with the electron and was there all along. Symmetrically, describing the electron's effect by the advanced field and subtracting it leaves the outgoing field.

Equation (13) makes the difference of F_{adv} and F_{ret} remarkably simple; we have

$$f^{\mu\nu} - F_{\text{in}}^{\mu\nu} = \frac{1}{2} (F_{\text{ret}}^{\mu\nu} - F_{\text{adv}}^{\mu\nu}) = \frac{e}{\kappa^{1/2}} \frac{2(\ddot{V}^\mu V^\nu - \ddot{V}^\nu V^\mu)}{3} \quad \longrightarrow \quad \frac{2e(\ddot{V}^\mu V^\nu - \ddot{V}^\nu V^\mu)}{3} \quad \text{as } \epsilon \rightarrow 0$$

If we regard the f in (4) as the electron's net field and F_{in} as the external field, then the Lorentz force produced by this averaged field,

$$\frac{e(F_{\text{ret}}^{\mu\nu} - F_{\text{adv}}^{\mu\nu})}{2} V_\nu = \frac{2e^2}{3} (\ddot{V}^\mu + \dot{V}^2 V^\mu)$$

is precisely the electron's self radiation-reaction force—a correction to the Abraham–Lorentz theory, and in complete agreement with the result of Jackson problem 16.7.